# Extrauterine Pregnancy 05. hCG: Powerful, Misused, and Never Alone

Companion reading handout for simultaneous listening.
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## Transcript

Extrauterine Pregnancy audio course. Episode 5. hCG: Powerful, Misused, and Never Alone.

hCG is a trophoblast signal measured over time. It is not a location test.

That distinction is simple, but it prevents several of the most consequential errors in early pregnancy care. hCG can help interpret whether trophoblast is rising, falling, plateauing, or responding to treatment. It can help decide when ultrasound should be repeated. It can help identify a pregnancy that is behaving abnormally. It cannot, by itself, tell you whether the pregnancy is in the endometrial cavity, the tube, the cervix, a cesarean scar, the ovary, or the abdomen.

The discriminatory zone was created to connect a blood value with an ultrasound expectation. Kadar's original work used transabdominal ultrasound and proposed a range above which a gestational sac should be seen in a normal intrauterine pregnancy. As ultrasound improved and transvaginal scanning became standard, reported discriminatory levels dropped, often into the range of 1000 to 2000 milli-international units per milliliter. The concept was clinically attractive: if hCG is high enough and the uterus is empty, the pregnancy must be abnormal.

The concept is useful as a probability tool. It becomes unsafe when it is used as an irreversible-treatment trigger.

Doubilet and Benson tested the exact danger. They identified 202 patients whose transvaginal ultrasound showed no intrauterine fluid collection on the same day as a positive beta hCG, but who later had a live intrauterine pregnancy documented. Nine of those patients had hCG values at or above 2000. The highest hCG was 6567, and the highest value preceding a term liveborn baby was 4336. Their conclusion was direct: the hCG discriminatory level should not determine management in a hemodynamically stable patient with suspected ectopic pregnancy when ultrasound shows neither intrauterine nor ectopic pregnancy.

Those numbers should slow the resident down. Nine patients above 2000 means the traditional comfort of "above the zone" was not safe. A value of 6567 before a later live intrauterine pregnancy means even a very high hCG can be earlier than the visible ultrasound anatomy in a real patient. The term liveborn case after 4336 is the number that should sit in memory whenever methotrexate is being considered in a stable desired pregnancy with a nondiagnostic scan. The number is not telling you that ectopic is unlikely. It is telling you that irreversible treatment cannot rest on hCG alone.

This is not a minor technical caveat. It means that a stable patient with hCG above a chosen threshold and no visible intrauterine sac may have ectopic pregnancy, early pregnancy loss, or a developing intrauterine pregnancy that is not yet visible. If methotrexate or uterine evacuation is performed before those possibilities are reasonably separated, a desired viable pregnancy can be harmed.

ACOG therefore advises that hCG values alone should not be used to diagnose ectopic pregnancy and should be correlated with history, symptoms, and ultrasound. If the discriminatory concept is used, ACOG recommends using a conservatively high value, as high as 3500 milli-international units per milliliter, to reduce the risk of interrupting a desired intrauterine pregnancy.

The 3500 value should be heard as a safety buffer, not as a permission line. A lower discriminatory threshold tries to make diagnosis earlier, but it increases the risk of misclassifying a normally sited pregnancy that is not yet visible. A higher threshold sacrifices some diagnostic impatience to protect a desired intrauterine pregnancy. The threshold is therefore not a biological wall. It is a clinical compromise between the danger of missing ectopic pregnancy and the danger of treating a pregnancy that could have continued.

The resident should understand why the threshold varies. hCG production and visible anatomy usually develop in parallel, but not perfectly. Gestational dating may be wrong. Ovulation may have occurred later than expected. Multiple gestation can produce higher hCG before ultrasound landmarks appear. Fibroids, uterine orientation, maternal body habitus, equipment resolution, scan route, and operator expertise affect what is seen. Even the definition of what counts as a true gestational sac matters, because a nonspecific intrauterine fluid collection is not the same as a sac with yolk sac or embryo.

Serial hCG is different from a single value because it introduces time.

The older rule that hCG should double every 48 hours was too rigid. Barnhart's work on symptomatic patients with early viable intrauterine pregnancies showed that viable pregnancies can rise more slowly than older teaching allowed, and that the expected minimal rise depends on the starting value. ACOG summarizes the conservative thresholds as approximately 49 percent rise over 48 hours when initial hCG is below 1500, 40 percent when the initial value is 1500 to 3000, and 33 percent when the initial value is above 3000.

Teach the sequence as a curve flattening, not as three disconnected facts. Below 1500, the earliest viable pregnancies are expected to rise faster, so the conservative minimum is about 49 percent. Between 1500 and 3000, the rise can be slower, about 40 percent. Above 3000, the curve slows further, about 33 percent. The larger the starting signal, the less dramatic the next proportional rise has to be. That is why the old "doubling" rule was too blunt.

These numbers should not be memorized as three isolated cutoffs. They teach the shape of early hCG biology: rise slows as the starting concentration gets higher. A slower-than-expected rise is concerning for abnormal pregnancy, including ectopic pregnancy or early loss, but it is not specific enough to locate the pregnancy or prove nonviability by itself.

Falling hCG has a similar limitation.

Barnhart's decline study examined patients ultimately diagnosed with spontaneous abortion and showed that the expected decline depends on the starting hCG. The slowest expected decline over 48 hours ranged roughly from 21 to 35 percent, and by 7 days from 60 to 84 percent, depending on initial level. That work helps identify a decline that is too slow for a typical resolving miscarriage and should prompt concern for ectopic pregnancy or retained trophoblast.

Here again, make the number meaningful. A falling hCG is supposed to fall with a certain speed when an intrauterine loss is resolving. If it falls too slowly, the body may still contain active trophoblast somewhere: ectopic tissue, retained tissue, or another abnormal source. The 21 to 35 percent at 48 hours and 60 to 84 percent by 7 days are not location rules. They are tempo rules. They ask whether the decline is fast enough for ordinary resolution.

But even an appropriate decline does not prove location. Barnhart explicitly notes that a decrease within expected parameters does not rule out rupture of a resolving ectopic pregnancy. Novak also warns that tubal rupture can occur even when beta hCG levels are falling. A falling value is improving trophoblastic activity; it is not a signed certificate that the pregnancy was intrauterine.

This is why hCG interpretation must remain attached to symptoms and ultrasound. Severe pain, syncope, shoulder-tip pain, peritoneal signs, or increasing hemoperitoneum can outrank a reassuring curve. A definite ectopic on ultrasound can outrank an hCG pattern. A stable desired pregnancy with nondiagnostic ultrasound deserves protection from premature irreversible treatment.

Heterotopic pregnancy exposes another limitation. If a normally sited pregnancy and an ectopic pregnancy coexist, the normally sited pregnancy can drive the hCG pattern. The hCG may rise appropriately while the ectopic pregnancy remains dangerous. This is especially relevant after IVF or ovulation induction, where the pretest probability is higher than in spontaneous conception.

Finally, hCG can mislead for reasons outside ordinary ectopic diagnosis. Gestational trophoblastic neoplasia can follow any gestational event, including ectopic pregnancy, and hCG is central to diagnosis and follow-up. False-positive serum hCG can occur from assay interference such as heterophile antibodies. A persistently positive serum hCG with a negative urine hCG should raise the possibility of phantom hCG, because heterophile antibodies generally do not pass into urine. Quiescent gestational trophoblastic disease can produce persistent low real hCG, often below 500 milli-international units per milliliter, for weeks to months.

These are not routine explanations for every PUL. They are reminders that the number is a measurement of a biological signal, and the clinician still has to decide what tissue is producing that signal.

The practical hCG discipline is this.

Use the value only after naming the clinical state. Is the patient stable? Has location been demonstrated? Is the pregnancy desired? What did ultrasound actually show? What was the exact time interval between values? Is the change a rise, fall, or plateau? Does the magnitude of change fit the patient's symptoms and scan?

If those answers do not align, do not average them into false reassurance. Reassess the patient. Repeat imaging when appropriate. Involve senior review. Protect the patient from rupture, and protect a possible desired intrauterine pregnancy from treatment given before the evidence is strong enough.

Use discordance as a structured bedside drill. First, ask whether the patient is unstable; if she is, physiology outranks the spreadsheet. Second, ask whether a normally sited pregnancy has truly been proven; an empty uterus, a nonspecific fluid collection, or a pregnancy in a scar or cervix is not that proof. Third, ask whether ectopic pregnancy is definite, probable, possible, or merely not yet visible. Fourth, ask whether the plan is safe until the next data point. A stable patient with uncertainty can wait under surveillance. An unstable patient cannot. A definite ectopic does not need hCG permission to be treated. A desired pregnancy with a nondiagnostic scan should not be destroyed by a threshold alone.

This is the mental order when the story, hCG, and ultrasound disagree: danger first, diagnostic certainty second, reversibility third. Do not average the data. Rank them. Pain, syncope, peritoneal signs, and hemoperitoneum can make the case urgent. A yolk sac or fetal pole outside the normal site can make the diagnosis. Uncertainty in a stable desired pregnancy should preserve options until the next reliable measurement or expert scan. That ordering is what keeps the resident from both fatal delay and premature irreversible treatment.

hCG is powerful because it describes trophoblast through time. It is dangerous when clinicians ask it to describe place.